Precipitation

The water or ice particles reaching the ground surface from the atmosphere is known as precipitation. The various forms of precipitation are dew, mist, fog, sleet, drizzle, rain and hail. Conventionally, the amount of rain has been recorded by the instrument called Rain gauge. The amount of rain will be recorded in mm or cm or inches at a given time. The Automated Weather Station (AWS) instrument records the rainfall with the help of sensors.

Sunshine

Sunshine means that the sun's rays lie within the visible spectrum that reaches the earth's surface. The amount or duration of sunshine in hours per day at a given region is measured using a sunshine recorder. The sunshine of a place depends on various seasons.

Visibility

Visibility means the transparency of the air in the particular place. It depends on the presence of water, ice, dust and smoke particles in the atmosphere. Visibility is measured by the distance at which prominent objects can be seen and the details discerned. The scale of visibility varies from zero, when objects cannot be easily seen beyond 25 metres, to visibility 9, when objects can be easily seen at a distance of 50 kilometres. Scale 0 to 3 indicate Fog, scale 4 represents Haze or Mist, scale 5 to 9 represents poor to Excellent visibility.

12.3 Advancement in Measuring Weather Elements

In recent days, weather stations use weather balloons, aircraft and ships to carry the instruments for measuring the temperature, pressure, humidity in the lower atmosphere. Weather Radars are used to locate precipitation, type, intensity.

Radiosonde (an instrument carried by balloon or other means to various levels of the atmosphere and transmitting measurements by radio) is an instrument used to measure the vertical profiles of temperature, dew point and winds. It estimates the wind direction, velocity based on radar tracking.

Weather satellites help in providing all weather information accurately. Some of the important weather satellites are: METSAT-1/KALPANA-1, OCEANSAT-2, INSAT -3D, SCATSAT-1

12.4 Weather Symbols

Symbols are the pictorial representation of the various elements. On a weather map, weather elements are represented using symbols. Weather codes are used in weather charts as meteorological symbols. Weather symbols are created and standardized by the World Meteorological organization and also by Natural Weather Bureaus. Knowledge of weather symbols is prime for weather map interpretation and weather forecasting. Weather symbols are listed below for precipitation, wind direction, cloud cover and sea conditions. Weather symbols depicted in weather charts help in weather forecasting.

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